

Variables and Controls

Identify every variable before you start. Uncontrolled variables are the usual reason a result cannot be trusted.

Name _____ Date _____

Worked example

Question: How does salt concentration affect the melting time of ice?

Independent: salt concentration (0, 5, 10, 15 g per 100 ml).

Dependent: time to melt (s), measured with a stopwatch.

Controlled: ice mass, starting temperature, container, room temperature.

Control group: ice in pure water (0 g of salt).

My investigation question

Independent variable What you change, and the range of values you will use.

Dependent variable What you measure, with units and the instrument used.

Controlled variables What you hold constant, and how you will hold it constant.

Is my test fair? If someone changed something else as well, could you still trust the result?
